

A popular matrix $X^T X$

$y_i = \mu + b_1 x_{1i} + b_2 x_{2i} + e_i$

$y = Xb + e$

$X = \begin{bmatrix} 1 & x_{11} \\ 1 & x_{12} \\ \vdots & \vdots \\ 1 & x_{1n} \end{bmatrix}$

$X^T X = \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_{11} & x_{12} & \dots & x_{1n} \end{bmatrix} \times \begin{bmatrix} 1 & x_{11} \\ 1 & x_{12} \\ \vdots & \vdots \\ 1 & x_{1n} \end{bmatrix} = \begin{bmatrix} n & \sum_{i=1}^n x_{1i} \\ \sum_{i=1}^n x_{1i} & \sum_{i=1}^n x_{1i}^2 \end{bmatrix}$

(在线性模型分析中经常用到)

Another popular matrix $e^T e$

$e = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}$ 随机误差的向量

> Dot product or inner product (点积或内积)

$e^T e = [e_1 \ e_2 \ \dots \ e_n] \times \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix} = \sum_{i=1}^n e_i^2$ (标量)

> Outer product (外积) $ee^T = [c_{ij}]_{n \times n}$ (矩阵)

迹 trace 矩阵对角元素的和

> 矩阵的迹(trace of a matrix):

If $A_{n \times n} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$, then trace of matrix A is, $tr(A) = \sum_{i=1}^n a_{ii}$

(矩阵的对角元素的和)

Properties of trace:

$tr(A^T) = tr(A)$

$tr(A \pm B) = tr(A) \pm tr(B)$

$tr(AB) = tr(BA)$

(in general, $tr(AB) \neq tr(A)tr(B)$)

$tr(AA^T) = tr(A^T A) = \sum_{i,j} a_{ij}^2$

> 分割矩阵(partitioned matrices):

$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}$ row vector (向量)

$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$ column vector (向量)

$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ submatrix

※ Partitioned matrices can be used to simplify equations or to obtain new interpretation of equations (see the next slide)

含有 m 个方程和 n 个未知量的线性方程组(linear system)定义为:

$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$

$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$

$\dots$

$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$

$\Downarrow$

$Ax = b$

$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} = [c_1 \ c_2 \ \dots \ c_n]$ $n \times n$

$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ $n \times 1$

$b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$ $m \times 1$

Ax is a linear combination (线性组合) of the column vectors of matrix A (Ax 是矩阵 A 的列向量的线性组合):

$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} = [c_1 \ c_2 \ \dots \ c_n]$ and $x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

$\Rightarrow Ax = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \end{bmatrix} = x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \dots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$

$= x_1 c_1 + x_2 c_2 + \dots + x_n c_n \Rightarrow Ax$ can be viewed as the linear combination of the column vectors of A with coefficients (系数) $x_1, x_2, \dots, x_n$

$= [c_1 \ c_2 \ \dots \ c_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

← You can derive the same result if you perform the matrix multiplication for matrix A expressed in column vectors and x directly

> 线性独立 (linear independent)

If there only exists the zero vector $c=0$, to make $Ac=0$, then the column vectors of matrix A are called linear independent (矩阵 A 的列向量称为线性独立).

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> 矩阵的秩(rank of matrix)

The maximum number of vectors (rows or columns) which are linear independent in a matrix (矩阵相互线性独立的向量(列或行)数目, 称为矩阵的秩, 有列秩、行秩).

The rank of a matrix will not change, and is denoted by $\rho(A)$.

矩阵的秩用来度量矩阵的行向量(或列向量)之间的相关性。如果矩阵的行向量之间是线性无关的, 则矩阵就是行满秩的, 也就是秩等于行数。

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Rules on operation of matrix rank(矩阵的秩运算法则):

$$\rho(A) \leq \min(n, m)$$

$$\rho(AB) \leq \min\{\rho(A), \rho(B)\}$$

$$\rho(A+B) \leq \rho(A+B) \leq [\rho(A) + \rho(B)]$$

$$\rho(A) = \rho(A^T)$$

$$\rho(AA^T) = \rho(A^T A) = \rho(A) = \rho(A^T)$$

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> 幂等矩阵(idempotent matrix)

If $AA=A$ (A 为方阵), then A is idempotent matrix (幂等矩阵).

Properties of Idempotent matrix:

- 幂等矩阵的迹等于幂等矩阵的秩 $\text{tr}(A) = \rho(A)$
- K 次相乘仍等于最初的矩阵 A $A^k = A$

$I-A$ is also an idempotent matrix.

For example

$$AA=A = \begin{bmatrix} 0.5 & 0.5 & 0.0 \\ 0.5 & 0.5 & 0.0 \\ 0.0 & 0.0 & 1.0 \end{bmatrix} \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\rho(A) = 2 = \text{tr}(A)$$

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4. 矩阵求逆和广义逆矩阵 (inverse matrix and generalized inverse matrix)

> 逆矩阵(inverse matrix): If there exists a matrix A^{-1} , whose product with matrix A is a identity matrix (单位矩阵)

$$A^{-1}A = AA^{-1} = I$$

then A^{-1} is called the inverse matrix (逆矩阵) of matrix A .

For example,

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \quad A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}, \quad AA^{-1} = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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Rules on operation of inverse matrix (逆矩阵的运算法则):

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$(A^T)^{-1} = (A^{-1})^T$$

$$(A^{-1})^{-1} = A$$

$$D(d_{ii})^{-1} = D\left(\frac{1}{d_{ii}}\right)$$

$$(A + BCD)^{-1} = A^{-1} - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}DA^{-1}$$

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满秩矩阵(full rank matrix)

设A是 n 阶矩阵

If $\rho(A) = r = n$, then A is a full rank matrix (or non-singular matrix) (满秩矩阵), A^{-1} exists (可逆矩阵).

If $\rho(A) = r < n$, then there is no A^{-1} . A is called noninvertible (or singular) (不可逆矩阵或奇异矩阵).

满秩不局限于 n 阶矩阵。若矩阵秩等于行数，称为行满秩；若矩阵秩等于列数，称为列满秩。既是行满秩又是列满秩则为 n 阶矩阵即 n 阶方阵。行满秩矩阵就是行向量线性无关，列满秩矩阵就是列向量线性无关；所以如果是方阵，行满秩矩阵与列满秩矩阵是等价的。

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Gauss-Jordan method of matrix inversion:

$$[A \mid I] \xrightarrow{\text{Gauss-Jordan elimination}} [I \mid A^{-1}]$$

Find the inverse of the matrix A

$$A = \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix}$$

Solution:

逆矩阵

$$AX = I$$

$$\begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x_{11} + 4x_{21} & x_{12} + 4x_{22} \\ -x_{11} - 3x_{21} & -x_{12} - 3x_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

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by equating corresponding entries

$$\begin{aligned} \Rightarrow x_{11} + 4x_{21} &= 1 & (1) \\ -x_{11} - 3x_{21} &= 0 & (2) \end{aligned}$$

This two systems of linear equations share the same coefficient matrix (系数矩阵), which is exactly the matrix A

$$\begin{aligned} (1) \Rightarrow \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} x_{11} \\ x_{21} \end{bmatrix} &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow x_{11} = -3, x_{21} = 1 \\ (2) \Rightarrow \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} x_{12} \\ x_{22} \end{bmatrix} &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow x_{12} = -4, x_{22} = 1 \end{aligned}$$

Thus

$$X = A^{-1} = \begin{bmatrix} -3 & -4 \\ 1 & 1 \end{bmatrix}$$

Perform the Gauss-Jordan elimination on the matrix A with the same row operations

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Note:

Instead of solving the two systems separately, you can solve them simultaneously by appending (附加) the identity matrix to the right of the coefficient matrix

$$\begin{bmatrix} 1 & 4 & 1 & 0 \\ -1 & -3 & 0 & 1 \end{bmatrix} \xrightarrow{\text{Gauss-Jordan elimination}} \begin{bmatrix} 1 & 0 & -3 & -4 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

A I I A⁻¹

solution for $\begin{bmatrix} x_{11} \\ x_{21} \end{bmatrix}$ solution for $\begin{bmatrix} x_{12} \\ x_{22} \end{bmatrix}$

※ If A cannot be row reduced to I, then A is singular (奇异矩阵或不可逆矩阵)

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Ex. Find the inverse of the following matrix

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ -6 & 2 & 3 \end{bmatrix}$$

Solution:

$$[A \mid I] = \begin{bmatrix} 1 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ -6 & 2 & 3 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{R_2 - R_1} \begin{bmatrix} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ -6 & 2 & 3 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_3 + 6R_2} \begin{bmatrix} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & -4 & -3 & -6 & 6 & 1 \end{bmatrix}$$

$$\xrightarrow{R_1 + R_2} \begin{bmatrix} 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & -4 & -3 & -6 & 6 & 1 \end{bmatrix} \xrightarrow{R_3 + 4R_2} \begin{bmatrix} 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & -1 & -2 & 10 & 1 \end{bmatrix}$$

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$$\xrightarrow{R_3 \times (-1)} \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & -3 & -3 & -1 \\ 0 & 0 & 1 & -2 & -4 & -1 \end{bmatrix} \xrightarrow{R_1 + R_3} \begin{bmatrix} 1 & 0 & 0 & -2 & -4 & -1 \\ 0 & 1 & 0 & -3 & -3 & -1 \\ 0 & 0 & 1 & -2 & -4 & -1 \end{bmatrix}$$

$$= [I \mid A^{-1}]$$

So the matrix A is invertible, and its inverse is

$$A^{-1} = \begin{bmatrix} -2 & -3 & -1 \\ -3 & -3 & -1 \\ -2 & -4 & -1 \end{bmatrix}$$

Check it by yourselves:

$$AA^{-1} = A^{-1}A = I$$

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含有 m 个方程和 n 个未知量的线性方程组 (linear system) 定义为:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned}$$

$\Downarrow$

$$Ax = b$$

$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} = [c_1 \ c_2 \ \dots \ c_n]$
 $x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$
 $b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$

$Ax = b$
 $A^{-1}Ax = A^{-1}b$
 $x = A^{-1}b$

Rules of operations of inverse matrix

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$(A^T)^{-1} = (A^{-1})^T$$

$$(A^{-1})^{-1} = A$$

$$D(d_{ii})^{-1} = D\left(\frac{1}{d_{ii}}\right)$$

$$(A + BCD)^{-1} = A^{-1} - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}DA^{-1}$$

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

Proof:

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

伴随矩阵(adjoint matrix)
行列式(determinant)

Consider the product $A[\text{adj}(A)]$

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} C_{11} & \dots & C_{1n} \\ C_{21} & \dots & C_{2n} \\ \vdots & \vdots & \vdots \\ C_{m1} & \dots & C_{mn} \end{bmatrix} = \begin{bmatrix} \det(A) & 0 & \dots & 0 \\ 0 & \det(A) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \det(A) \end{bmatrix}$$

The entry at the position (i, j) of $A[\text{adj}(A)]$

$$a_{i1}C_{j1} + a_{i2}C_{j2} + \dots + a_{in}C_{jn} = \begin{cases} \det(A) & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

> 广义逆矩阵(generalized inverse matrix)

If $AA^{-}A = A$, then A^{-} is a **generalized inverse** (广义逆) of matrix A .
The number of the generalized inverse of matrix A is infinite.

For any matrix Z , $A^{-} + (I - A^{-}A)Z$ is also a generalized inverse of matrix A .

For example,

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \quad A^{-} = \begin{bmatrix} \frac{1}{6} & \frac{1}{6} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{6} & -\frac{1}{6} & \frac{2}{3} \end{bmatrix}, \quad A^{-}A = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

so $AA^{-}A = A$.

If $Z = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then $A^{-} + (I - A^{-}A)Z = \begin{bmatrix} \frac{1}{2} & \frac{1}{6} & \frac{1}{3} \\ 0 & \frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{6} & \frac{2}{3} \end{bmatrix}$

is also a generalized inverse of matrix A

> 正交矩阵(orthogonal matrices)

设 A 为一个 n 级实矩阵, 如果满足 $A^T A = A A^T = I$, 则称 A 是正交矩阵(orthogonal matrix), 通常用字母 Q 表示.

$$Q = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$Q^T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$QQ^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$Q^T = Q^{-1}$$

> 正规矩阵(Normal matrices)

设 A 为一个 n 级实矩阵, 如果满足 $A^T A = A A^T$, 则称 A 是正规矩阵(normal matrix),

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

$$AA^T = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} = A^T A$$

广义逆矩阵(generalized inverse matrix)

For any matrix that has $\rho(\underline{A}) = r$, we may get the following equation from singular value decomposition (奇异值分解)

$$\underline{A} = \underset{r \times r}{\underline{D}} \begin{bmatrix} \underline{D} & \underline{0} \\ \underline{0} & \underline{0} \end{bmatrix} \underset{r \times r}{\underline{Q}^T}$$

where $\underline{D}$ is a diagonal matrix (对角矩阵), $\underline{P}$ and $\underline{Q}$ are orthogonal matrix (正交矩阵):

$$\underline{P}^T \underline{P} = \underline{I}, \quad \underline{Q}^T \underline{Q} = \underline{I}$$

We can get a generalized inverse (广义逆矩阵) of matrix $\underline{A}$ by the following equation

$$\underline{A}^- = \underline{Q} \begin{bmatrix} \underline{D}^{-1} & \underline{E} \\ \underline{B} & \underline{C} \end{bmatrix} \underline{P}^T$$

where $\underline{B}, \underline{C}, \underline{E}$ are any matrices

Mose-Penrose 广义逆(Mose-Penrose generalized inverse)

对于任何矩阵, 都可以找到唯一的一个广义逆, 称为 Mose-Penrose 广义逆. 对于矩阵 $\underline{A}$, 其相应的 Mose-Penrose 广义逆以 $\underline{A}^+$ 表示.

M-P generalized inverse satisfies all following properties:

$$\underline{A}\underline{A}^+\underline{A} = \underline{A}, \quad \underline{A}^+\underline{A}\underline{A}^+ = \underline{A}^+, \quad (\underline{A}^+\underline{A})^T = (\underline{A}^+\underline{A}), \quad (\underline{A}\underline{A}^+)^T = (\underline{A}\underline{A}^+)$$

We can get $\underline{A}^+$ by $\underline{A}^+ = \underline{Q} \begin{bmatrix} \underline{D}^{-1} & \underline{0} \\ \underline{0} & \underline{0} \end{bmatrix} \underline{P}^T$

射影矩阵(projection matrix)

在线性模型分析中常用到射影矩阵(projection matrix).

For any matrix $\underline{X}$, we have $\underline{P}_X = \underline{X}(\underline{X}^T \underline{X})^{-1} \underline{X}^T$ which is the projection matrix of $\underline{X}$.

The properties of projection matrix $\underline{P}_X$:

$$\underline{P}_X \underline{X} = \underline{X}, \quad \underline{X}^T \underline{P}_X = \underline{X}^T, \quad \underline{P}_X^T = \underline{P}_X, \quad \underline{P}_X \underline{P}_X = \underline{P}_X$$

幂等矩阵

5. 随机向量和随机矩阵(random vector and random matrix)

常量矩阵(constant matrix): 如果矩阵或向量中的元素都是常量(constant), 则称为常量矩阵(constant matrix)或常量向量(constant vector).

随机矩阵(random matrix): 当矩阵或向量中的元素都是随机变量(random variable)时, 则称为随机矩阵(random matrix)或随机向量(random vector).

设 $\underline{x}, \underline{y}$ 为随机向量, $\underline{a}, \underline{b}, \underline{c}$ 等为常量向量; $\underline{X}, \underline{Y}$ 为随机矩阵, $\underline{A}, \underline{B}, \underline{C}$ 等为常量矩阵. 对随机矩阵或随机向量分别求期望值, 就是对其所有元素求期望值.

$$E(\underline{X}) = [E(x_{ij})]$$

$$E(\underline{x}) = [E(x_i)]$$

$$E(\underline{X}) = [E(x_{ij})]$$

$$E(\underline{x}) = [E(x_i)]$$

常量矩阵或常量向量的期望值就是原矩阵或原向量,

$$E(\underline{A}) = \underline{A}$$

$$E(\underline{a}) = \underline{a}$$

随机矩阵 (随机向量) 以下是对矩阵或向量的函数求期望值,

$$\left\{ \begin{array}{l} E(\underline{X} \pm \underline{Y}) = E(\underline{X}) \pm E(\underline{Y}) \\ E(\underline{A}\underline{X}) = \underline{A}E(\underline{X}) \\ E(\underline{X}\underline{B}) = E(\underline{X})\underline{B} \\ E(\underline{A}\underline{X} \pm \underline{Y}) = \underline{A}E(\underline{X}) \pm E(\underline{Y}) \\ E(\underline{A}\underline{X} \pm \underline{B}\underline{Y}) = \underline{A}E(\underline{X}) \pm \underline{B}E(\underline{Y}) \end{array} \right.$$

在多变量分析中, 经常需对随机向量求方差或协方差.

对随机向量 $\underline{x}$ 求方差定义为:

$$\text{Var}(\underline{x}) = E\{[\underline{x} - E(\underline{x})][\underline{x} - E(\underline{x})]^T\}$$

$$= [\sigma_{ij}]$$

$$= \underline{V} \text{ 或 } \underline{V}_x$$

其中 $\sigma_{ii} = \sigma_i^2 = \text{Var}(x_i)$ 是变量 x_i 的方差, $\sigma_{ij} = \sigma_{ji} = \text{Cov}(x_i, x_j) = \text{Cov}(x_j, x_i)$ 是变量 x_i 与 x_j 的协方差 ($i \neq j$).

对随机向量 $\underline{x}$ 和 $\underline{y}$ 求协方差定义为:

$$\text{Cov}(\underline{x}, \underline{y}) = E\{[\underline{x} - E(\underline{x})][\underline{y} - E(\underline{y})]^T\}$$

$$= [\sigma_{ij(xy)}]$$

$$= \underline{C}_{xy} \text{ 或 } \underline{V}_{xy}$$

其中 $\sigma_{ij} = \text{Cov}(x_i, y_j)$ 是变量 x_i 和 y_j 的协方差.

$$\text{Var}(\underline{X}) = \begin{bmatrix} \sigma_{11}^2 & \sigma_{12} & \dots & \sigma_{1n} \\ \sigma_{21} & \sigma_{22}^2 & \dots & \sigma_{2n} \\ \dots & \dots & \dots & \dots \\ \sigma_{n1} & \sigma_{n2} & \dots & \sigma_{nn}^2 \end{bmatrix}$$

$$\text{Var}(\underline{XY}) = \begin{bmatrix} \sigma_{11}(XY) & \sigma_{12}(XY) & \dots & \sigma_{1n}(XY) \\ \sigma_{21}(XY) & \sigma_{22}(XY) & \dots & \sigma_{2n}(XY) \\ \dots & \dots & \dots & \dots \\ \sigma_{n1}(XY) & \sigma_{n2}(XY) & \dots & \sigma_{nn}(XY) \end{bmatrix}$$

上述的V是对称矩阵。常量向量的方差和协方差均为零。

对随机向量的函数求方差或协方差的基本运算法则：

$$\begin{aligned} \text{Var}(y \pm a) &= \text{Var}(y) \\ \text{Var}(a^T y) &= a^T V a \\ \text{Var}(Ay) &= A V A^T \\ \text{Var}(x \pm y) &= \text{Var}(x) + \text{Var}(y) \pm \text{Cov}(x, y) \pm \text{Cov}(y, x) \\ &= V_X + V_Y \pm C_{XY} \pm C_{YX} \\ \text{Cov}(a^T x, b^T y) &= a^T C_{XY} b \\ \text{Cov}(Ax, y^T B^T) &= A C_{XY} B^T \\ \text{Cov}(x, a^T \pm y^T B^T) &= \pm C_{XY} B^T \end{aligned}$$

$$\text{Var}(Ay) = A V A^T$$

6. 二次型的分布(distribution of quadratic form)

二次型(quadratic form): n个变量的二次多项式称为二次型, 即在一个多项式中, 未知数的个数为任意多个, 但每一项的次数都为2的多项式。

在多元回归分析及方差分析中常用二次型(quadratic form)表示平方和。设随机变量x具有多变量正态分布(multivariable normal distribution)

$$x \sim MVN(\mu_X, V_X)$$

则

$$x^T A x = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j$$

是随机变量x的二次型(quadratic form), 其中A是对称常量矩阵。

An example of quadratic form

The quadratic form $3x_1^2 + x_2^2 + 2x_3^2 + 4x_1x_2 + 5x_1x_3 - 6x_2x_3$ can be expressed as,

$$3x_1^2 + x_2^2 + 2x_3^2 + 4x_1x_2 + 5x_1x_3 - 6x_2x_3 = x^T A x$$

where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and $A = \begin{bmatrix} 3 & 4 & 5 \\ 0 & 1 & -6 \\ 0 & 0 & 2 \end{bmatrix}$

In general, any quadratic form $x^T A x$ can be expressed as,

$$x^T A x = x^T \left(\frac{A + A^T}{2} \right) x, \quad \left(\frac{A + A^T}{2} \right) = \begin{bmatrix} 3 & 2 & 5/2 \\ 2 & 1 & -3 \\ 5/2 & -3 & 2 \end{bmatrix}$$

$$X X^T = \begin{bmatrix} x_1^2 & x_1 x_2 & x_1 x_3 \\ x_2 x_1 & x_2^2 & x_2 x_3 \\ x_3 x_1 & x_3 x_2 & x_3^2 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} (x_1 \ x_2 \ x_3)$$

$x^T A y$ 是随机变量x和y的二次型 (quadratic form). 常用来表示协方差分析(covariance analysis)的叉积和(sum of cross product).

二次型 $x^T A x$ 的均值和方差分别为:

$$\begin{cases} E(x^T A x) = \text{tr}(A V_X) + \mu_X^T A \mu_X \\ E(x^T A y) = \text{tr}(A C_{XY}) + \mu_X^T A \mu_Y \\ \text{Var}(x^T A x) = 2 \text{tr}(A V_X)^2 + 4 \mu_X^T A V_X A \mu_X \\ \text{Cov}(x, x^T A x) = 2 V_X A \mu_X \\ \text{Cov}(x^T A x, x^T B x) = 2 \text{tr}(A V_X B V_X) + 4 \mu_X^T A V_X B \mu_X \end{cases}$$

We often need to identify the independence of square sums, that is, the independence of quadratic form of random variables in multivariate regression analysis and variance analysis.

在多元回归分析及方差分析中经常需要辨别平方和的独立性, 即随机变量的二次型独立性。

如果 $A V_X B = 0$, 那么 $x^T A x$ 和 $x^T B x$ 相互独立。